

Project Description for Portfolio

Project Title: Nation Work

Overview: Nation Work is a comprehensive, full-stack freelance and gig marketplace platform designed to seamlessly connect clients with skilled service providers. Built with a high-performance backend, the platform manages the entire lifecycle of freelance engagements—from job posting and bidding (candidatures) to real-time communication, scheduling, and secure payments.

Key Features:

- **Job Posting & Bidding (Candidatures):** Clients can post service requests while freelancers can submit tailored proposals, including proposed prices, estimated timelines, and cover letters.
- **Real-Time Communication:** An integrated live chat system allowing clients and freelancers to communicate instantly and negotiate terms directly on the platform.
- **Scheduling & Calendar:** Built-in calendar management for booking services, setting milestones, and tracking project deadlines.
- **Trust & Verification (KYC):** A robust user verification system handling identity checks and KYC (Know Your Customer) flows to ensure a trusted marketplace environment.
- **Reviews & Evaluations:** A two-way rating and evaluation system enabling users to build public reputations based on past project performance.
- **Secure File Management:** Fast and secure handling of portfolio assets, document attachments, and avatars using S3-compatible object storage.
- **Dashboard & Analytics:** Comprehensive overview for users to track their active jobs, pending proposals, financial earnings, and overall statistics.

Technologies Used

Frontend

- **Framework:** React.js (A modern, component-based UI for seamless user interactions and dynamic state management).

Backend (High-Performance API)

Built for speed, concurrency, and reliability, utilizing a modern Python stack.

- **Core Framework:** FastAPI & Python 3.10+ (Running on Uvicorn ASGI server for ultra-fast asynchronous request handling).
- **Relational Database:** PostgreSQL with SQLAlchemy (ORM for complex relational data modeling between users, services, bids, and reviews).
- **Database Migrations:** Alembic (For version-controlled schema updates).
- **Real-Time Chat:** WebSockets (Providing low-latency, persistent connections for the messaging system).
- **Authentication & Security:** JWT (JSON Web Tokens) with `python-jose`, `passlib`, and `bcrypt` for secure password hashing and role-based access control.
- **Object Storage:** MinIO (S3-compatible storage utilized for handling user uploads, profile pictures, and attachment files via presigned URLs).
- **Geolocation:** geopy (For location-based services, helping clients find local freelancers).

- **Data Validation: Pydantic** (For strict type-checking, request validation, and environment configuration).